

August 17, 2026

Dear CMG Hiring Team,

I am writing to apply for the Junior QA Engineer position with Computer Modelling Group in Calgary. With a Master of Science in Computer Science from the University of Calgary and hands-on experience in Python, C++, scientific computing, data validation, debugging, and Linux-based development, I am excited by the opportunity to help ensure the quality and reliability of CMG's advanced scientific software.

My graduate research focused on developing scalable software and data-management methods for large geospatial and Earth observation datasets. I built C++ and Python systems for processing, encoding, storing, and retrieving satellite imagery, and I worked carefully with performance, reproducibility, data quality, and validation. In one part of this work, I improved a 10-tensor encoding process from 233 seconds to 26 seconds using C++ multithreading, while tracking behavior across large scientific datasets.

This background fits CMG's need for someone who can validate scientific applications, reason about numerical behavior, and investigate defects with care. For my DEM super-resolution research, I developed Python and PyTorch workflows, evaluated models with terrain-specific metrics, and identified how random data splitting could produce overly optimistic results. I then designed region-based validation methods that gave a more realistic view of model performance. That experience strengthened my attention to test design, edge cases, reporting, and root-cause thinking.

I also bring backend and API experience. As a Back-End Developer at Asr Gooyesh Pardaz, I developed Django REST APIs for NLP, text-to-speech, and speech-to-text services backed by PostgreSQL, implemented authentication and payment-service logic, and contributed to a chatbot system using GPT-style responses and FAQ data. These responsibilities gave me practical experience testing API behavior, debugging service logic, and working with database-backed systems.

CMG's role is especially appealing to me because it combines Calgary-based work, scientific software, Python automation, performance and regression testing, CI/CD, Linux/HPC environments, and collaboration with developers and domain experts. While I have not worked specifically in reservoir simulation, my research background in scientific data systems, numerical validation, C++/Python development, and technical documentation is highly transferable.

I have also developed strong communication habits through teaching and research collaboration. As a teaching assistant at the University of Calgary and a Deep Learning Teaching Assistant with Neuromatch Academy, I explained technical concepts, reviewed project work, supported students through debugging and experimentation, and provided structured feedback. I enjoy careful, detail-oriented technical work and collaborating with others to improve software quality.

Thank you for considering my application. I look forward to the opportunity to discuss how my Python, C++, scientific computing, validation, and debugging background can contribute to CMG's software quality work.

Sincerely,
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